

HANDS-ON SESSION • 2 HOURS

Cybersecurity & Safe Browsing

Practical security hygiene, network safety, public Wi-Fi threat modeling, and defensive email practices.

SESSION INSTRUCTOR

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MODULES & PRACTICAL TOPICS

Secure Browsing

Browser hardening, privacy & DNS security

HTTP vs HTTPS

SSL/TLS handshakes, encryption & MITM risks

Wi-Fi & Public Wi-Fi Risks

Rogue APs, packet sniffing & safe VPN usage

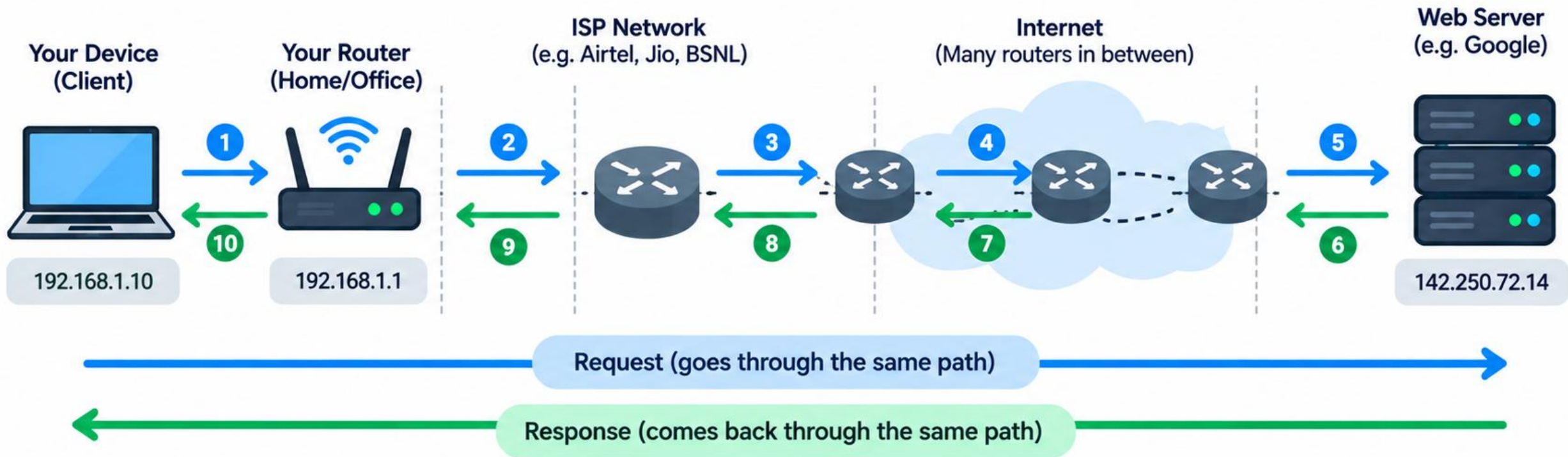
Email Safety & Phishing Labs

Header analysis, spoofing & real-world identification

Secure Browsing

Secure Browsing

How the Internet Works?



Secure Browsing

BROWSER



WEBSITE

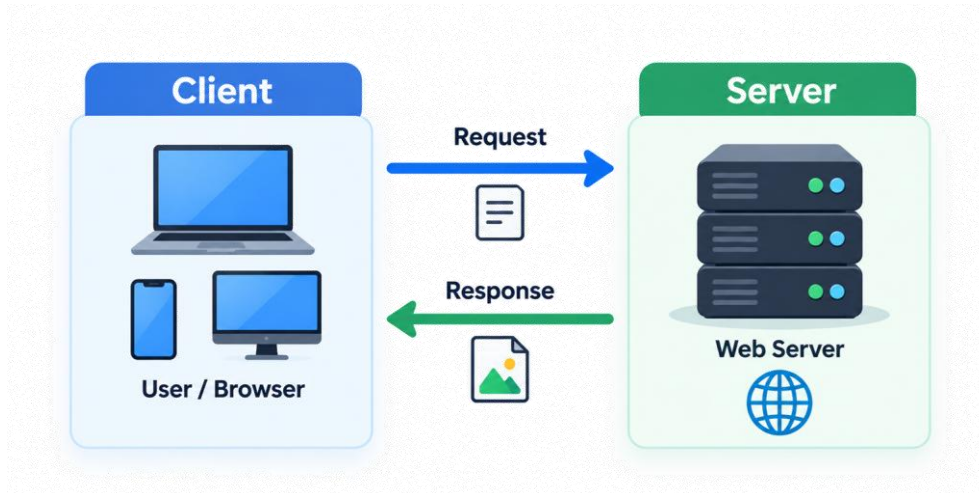


WHAT IS BROWSING?

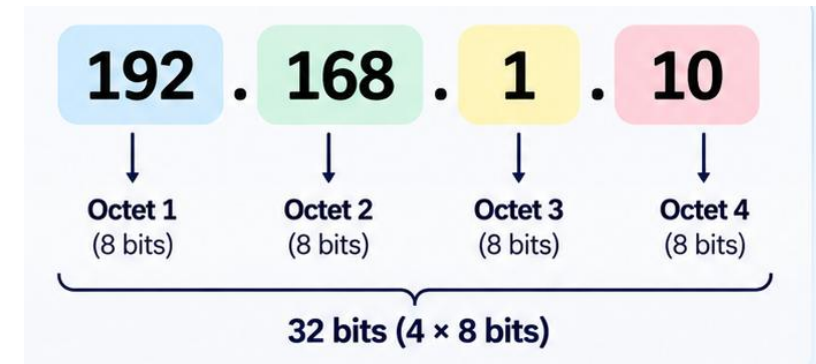


Secure Browsing

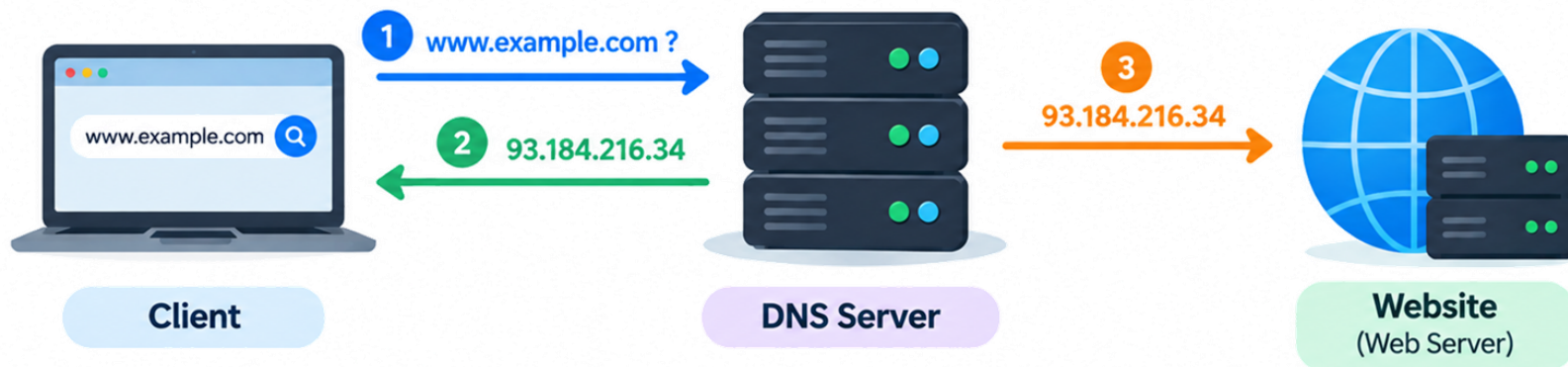
CLIENT ↔ SERVER



IP ADDRESS

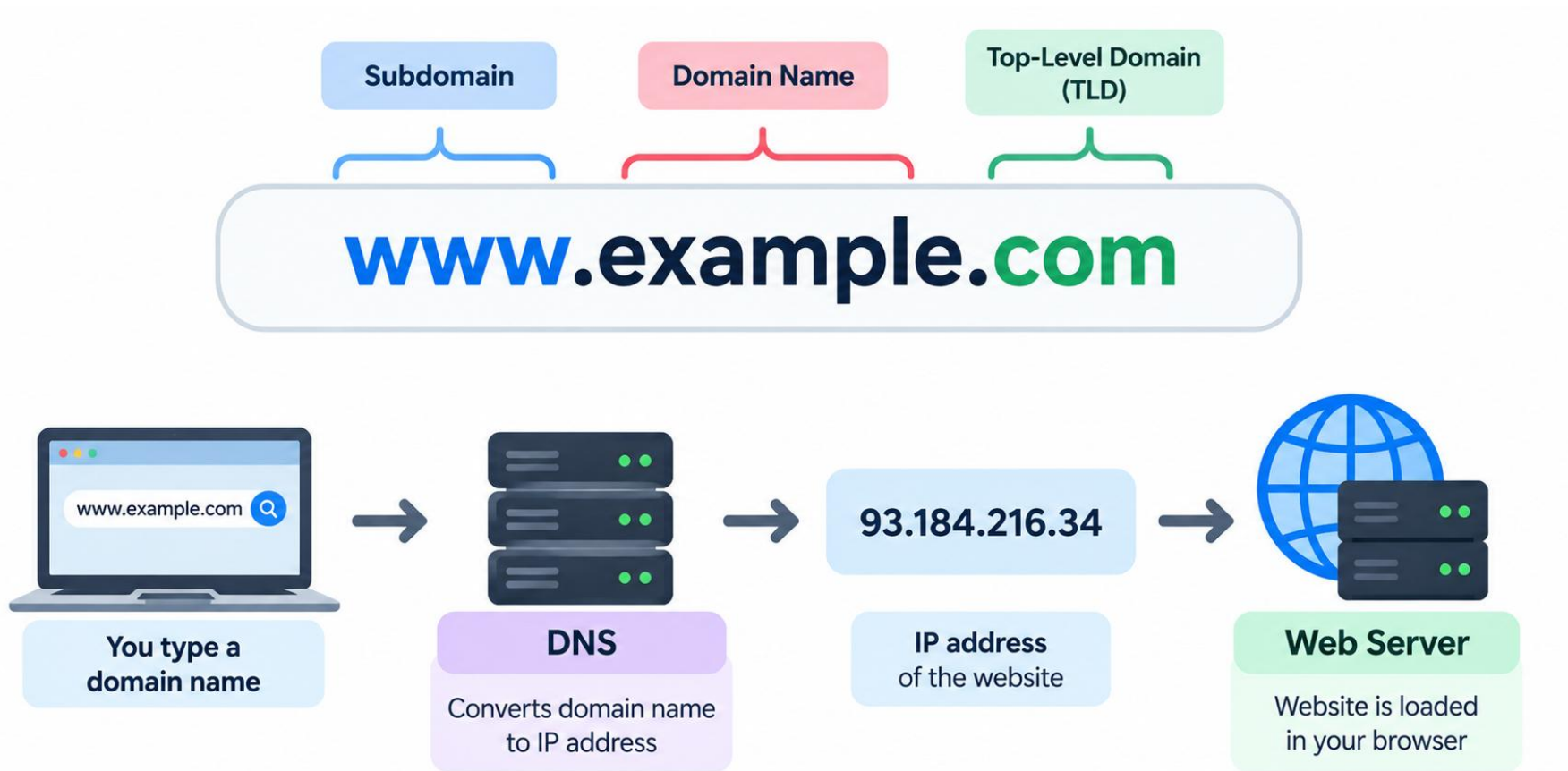


DNS



Secure Browsing

DOMAIN NAME



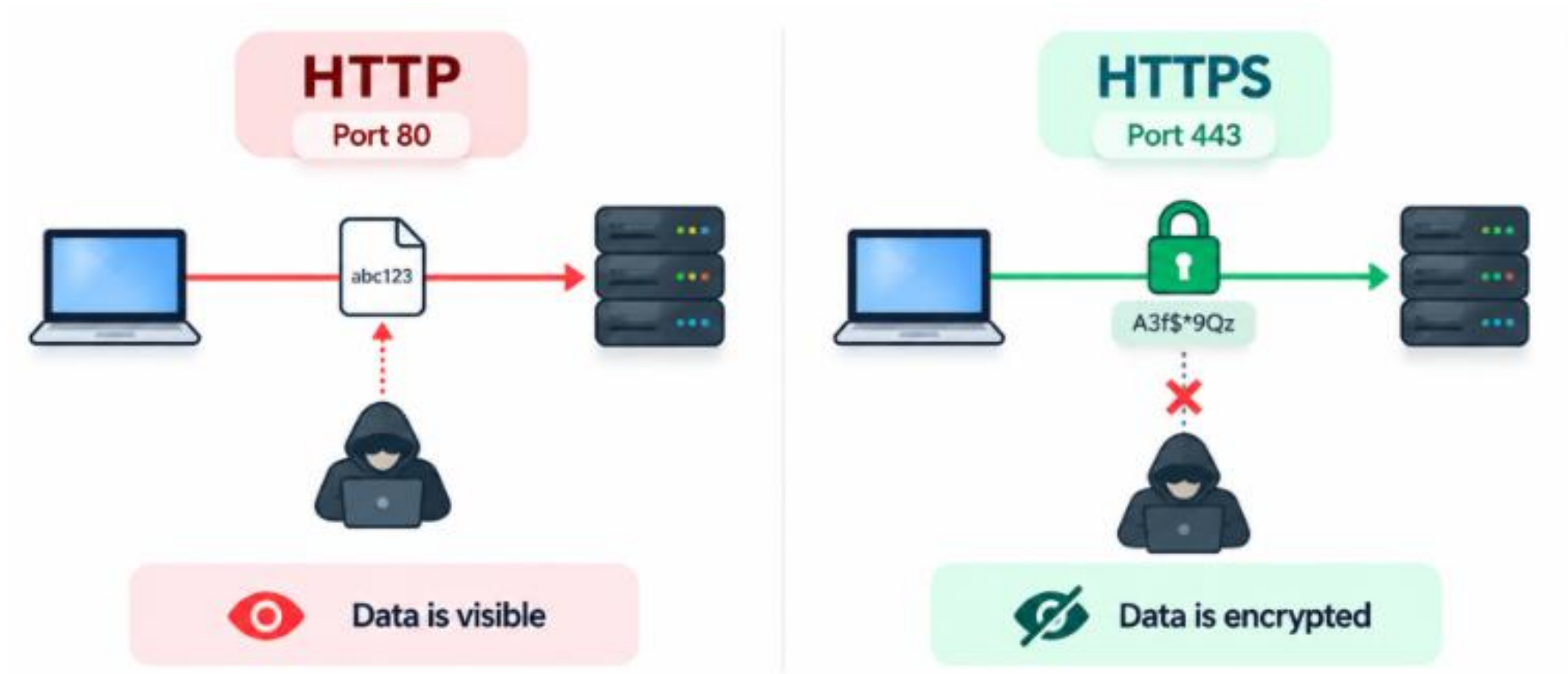
Secure Browsing

Demonstration

1. **ipconfig**
2. **ping**
3. **nslookup**
4. **tracert**
5. **Inspect dev tool**

HTTP vs HTTPS

HTTP vs HTTPS



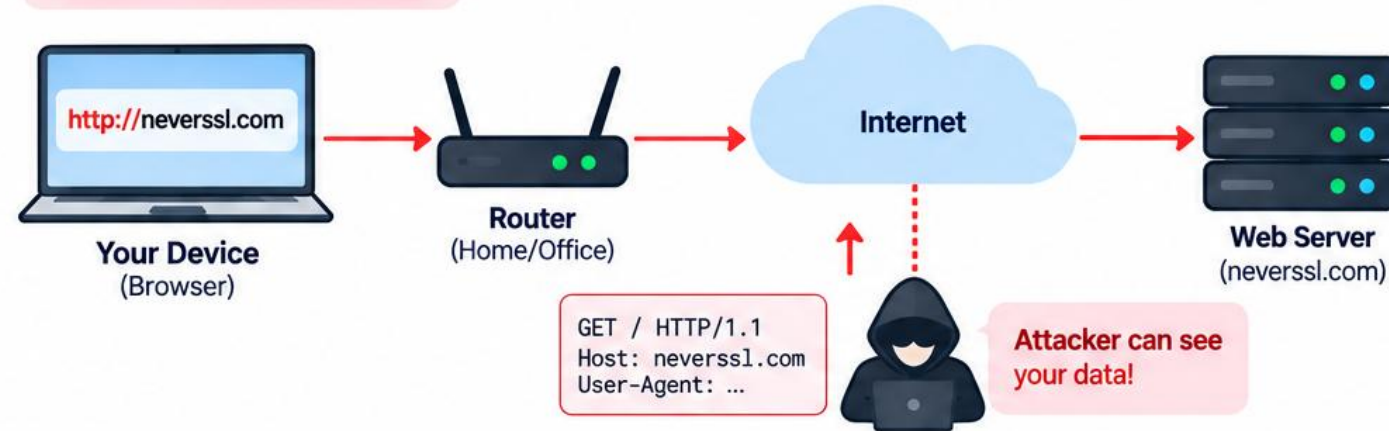
HTTP vs HTTPS

HTTP

(Port 80)

Plaintext Communication

Data is sent in readable form (not encrypted)

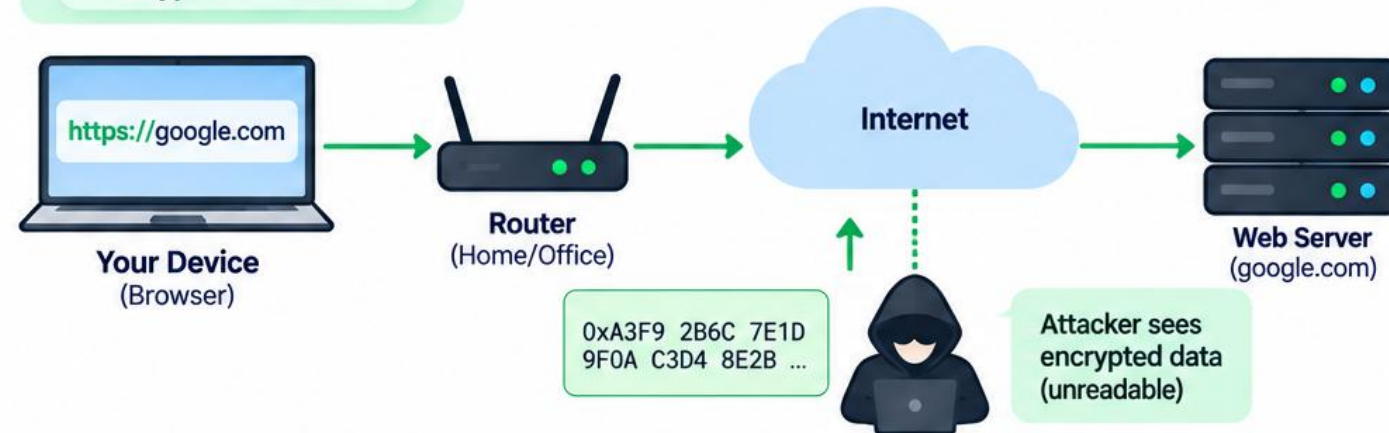


HTTPS

(Port 443)

Encryption in Transit

Data is encrypted using TLS (not readable)

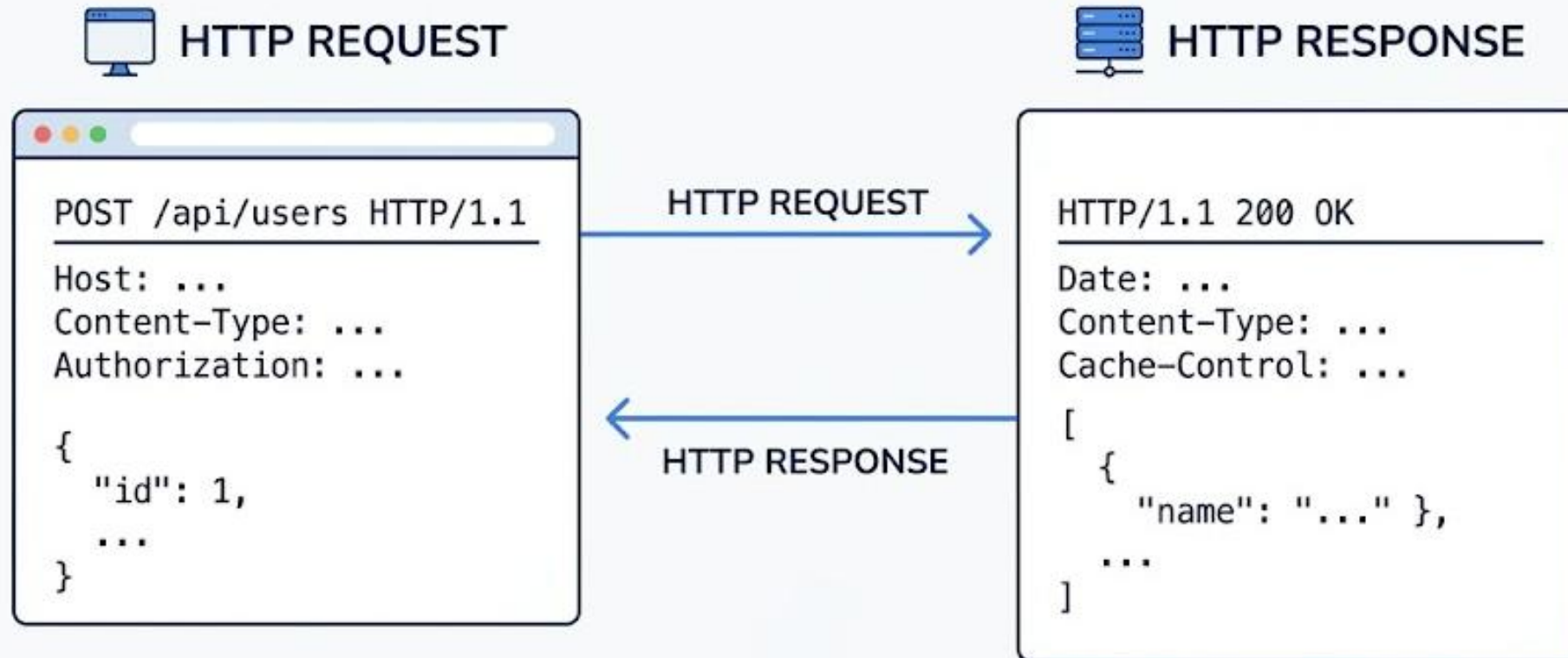


HTTP vs HTTPS

Demonstration

1. Wireshark

HTTP vs HTTPS



HTTP vs HTTPS

Demonstration

1. Inspect HTTP connection

HTTP vs HTTPS



SSL (Secure Sockets Layer)

SSL 2.0
1995

SSL 3.0
1996

 **Deprecated**
(Not Secure)



Older protocol
Designed in mid-1990s



Known security vulnerabilities
e.g., POODLE, BEAST



Less secure
Uses older cryptographic algorithms



Not recommended
Should not be used today



Timeline



Overview



Security



Algorithms



Current Status



TLS (Transport Layer Security)

TLS 1.0
1999

TLS 1.1
2006

TLS 1.2
2008

TLS 1.3
2018



Modern and improved version
Successor to SSL



Stronger security
Fixes SSL vulnerabilities



Uses stronger algorithms
e.g., AES, SHA-256, (Perfect Forward Secrecy)



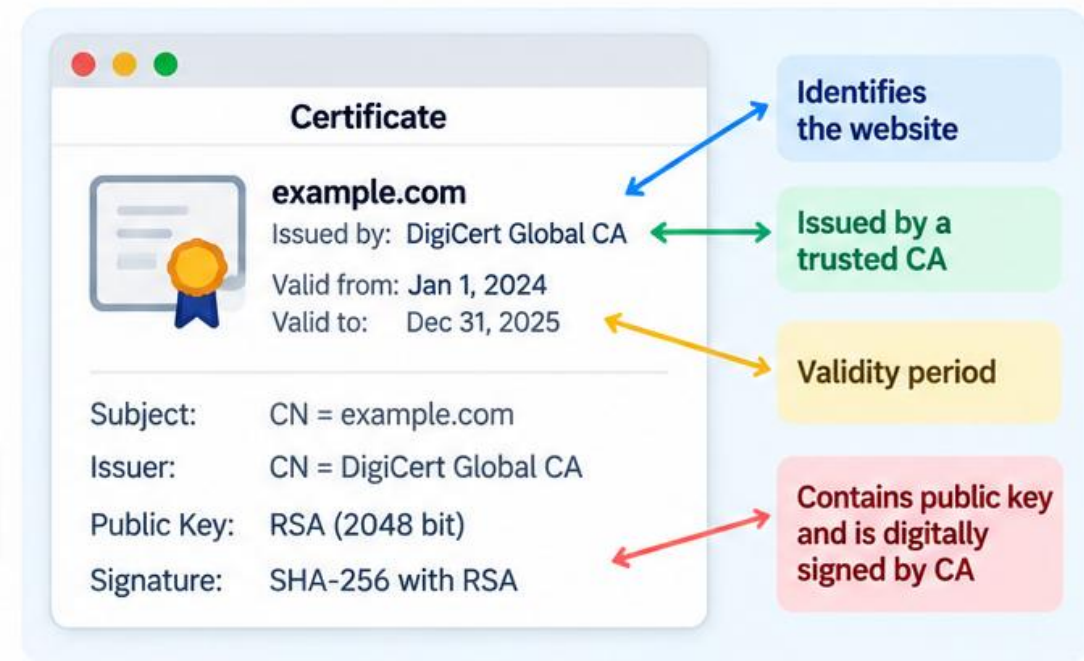
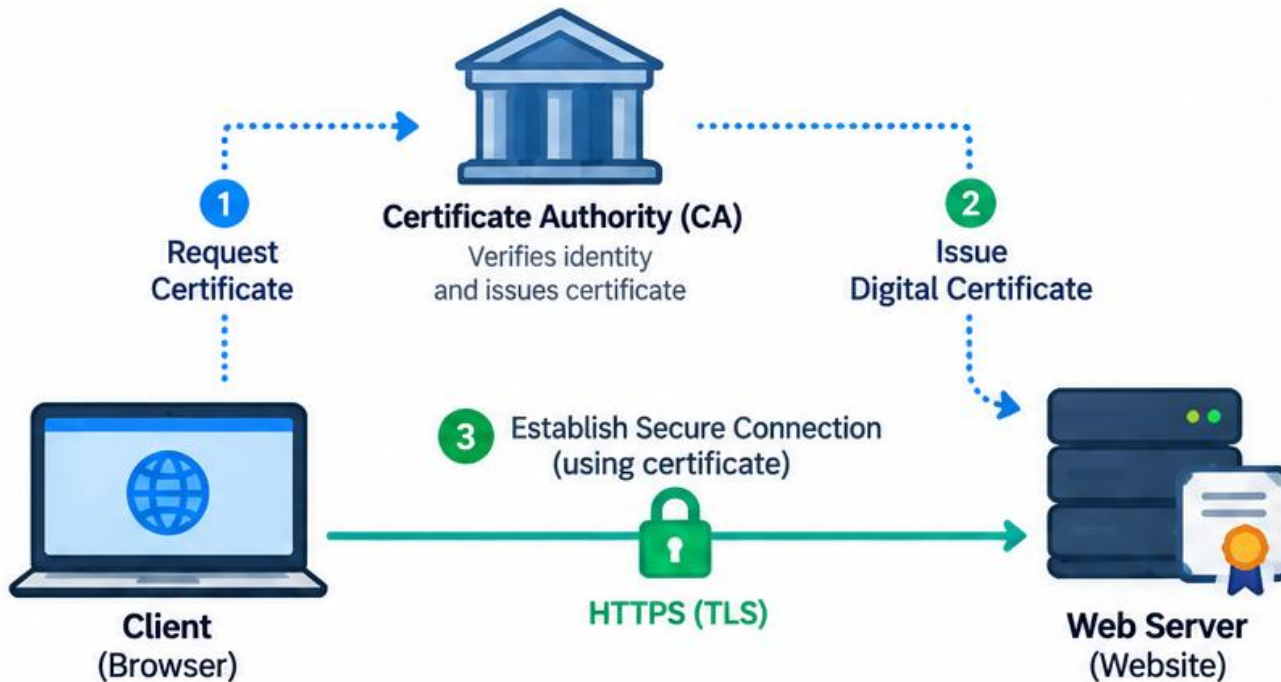
Widely used today
Recommended for secure communication

HTTP vs HTTPS



Digital Certificates

Prove identity. Enable trust. Power secure connections.



HTTP vs HTTPS

Demonstration

1. Browser Certificate

HTTP vs HTTPS

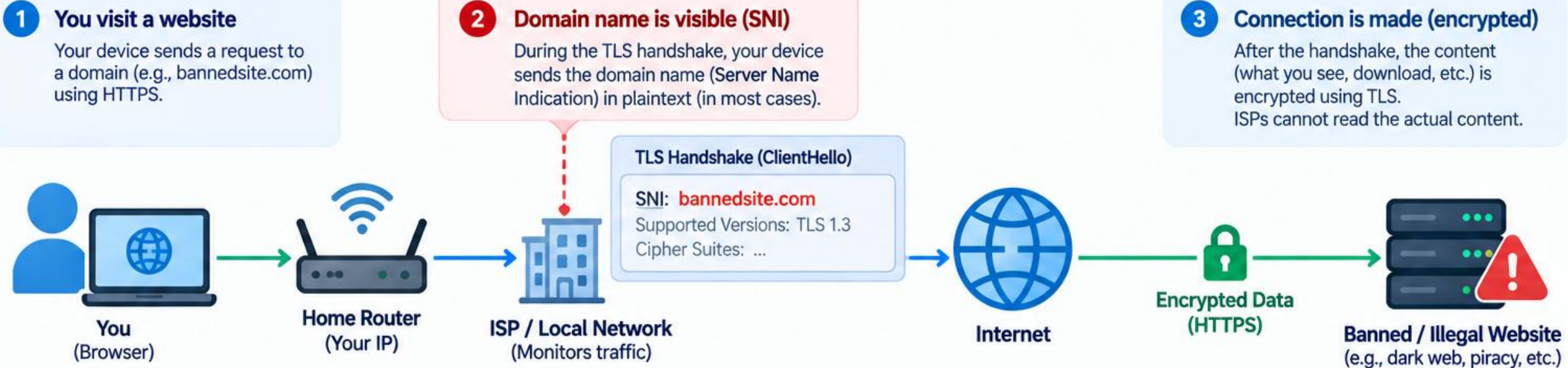
Why HTTPS matters

- **Stop Snoops:** Keeps private data (like passwords) hidden from eyes.
- **Block Tampering:** Stops hackers from changing what you send or see.
- **Verify Identity:** Confirms the website is real, not a fake scam site.
- **Boosts Trust:** Builds confidence and improves search rankings.

HTTP vs HTTPS

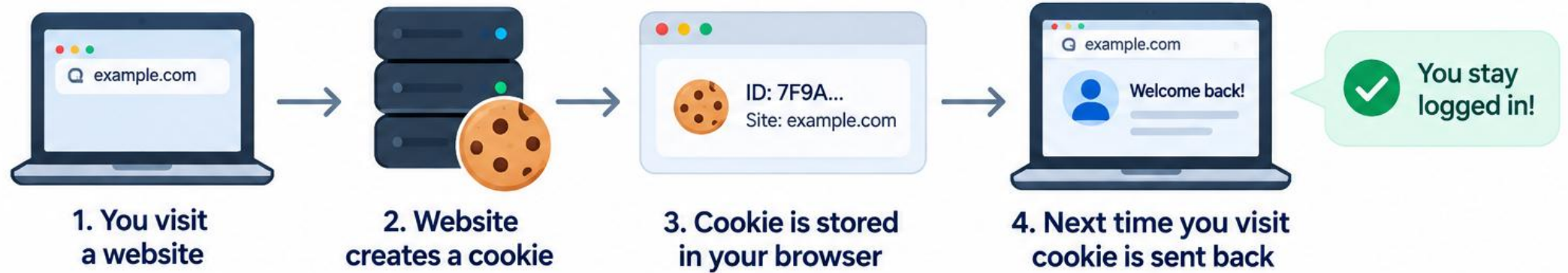
Why Governments Can Still Know You Access Banned Sites (Even with HTTPS)

HTTPS encrypts the content, but many other signals are still visible on the network.



Web Cookies

Web Cookies



Cookies – How Websites Remember You 🍪



🍪 Cookies don't contain your password, but they help websites remember you.

Web Cookies



Web Cookies

Demonstration

1. Insta Hack with cookie session id

Phishing & Social Engineering

Phishing & Social Engineering

What is phishing?

A **phishing attack** is a type of social engineering cyberattack where malicious actors send fraudulent messages designed to look like they come from a trusted person or organization.

Phishing & Social Engineering

Why Phishing works?

1 Psychological Triggers

Phishing messages use human emotions to make you act quickly.

Fear



Curiosity



Trust



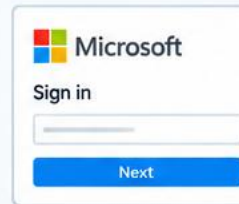
Anger / Revenge



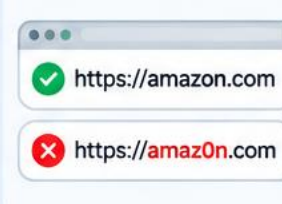
2 High-Quality Technical Mimicry

Phishing looks real because attackers can easily copy what you trust.

Visual Cloning



Lookalike Domains



AI-Generated Content



3 The Number Game (Low Cost, High Scale)

Attackers can send thousands (or millions) of fake messages with very little effort, and they only need a few people to fall for it.

Automation



Minimal Success Needed



Even if only a small percentage responds, it can still be very profitable for the attacker.

4 Overloaded Work Environments

People are busy, distracted and often access work on multiple devices, making them more likely to miss red flags.

Cognitive Overload



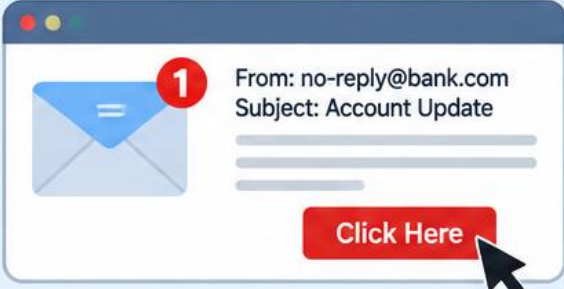
Mobile Devices



Phishing & Social Engineering

Types of Phishing

1 Email Phishing
Fake emails that look legitimate to trick you into clicking links, opening attachments or sharing information.



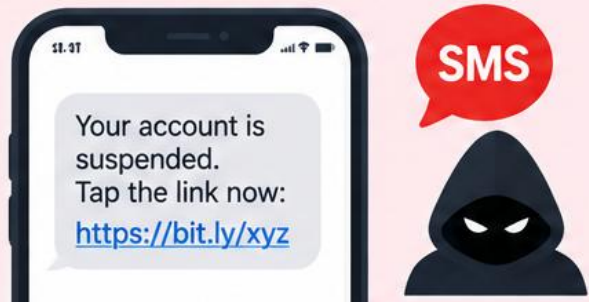
2 Spear Phishing
Targeted attacks on specific individuals (using personal information) to look more convincing.



3 Whaling
Targets high-profile individuals like CEOs, executives or senior officials for bigger gains.



4 Smishing
Phishing via SMS (text messages) that trick you into clicking a link or sharing information.



5 Vishing
Phishing via voice calls, where attackers pretend to be from a trusted organization to steal information.



6 QR Phishing (Quishing)
Uses malicious QR codes that, when scanned, lead you to fake websites or download malicious content.



Phishing & Social Engineering

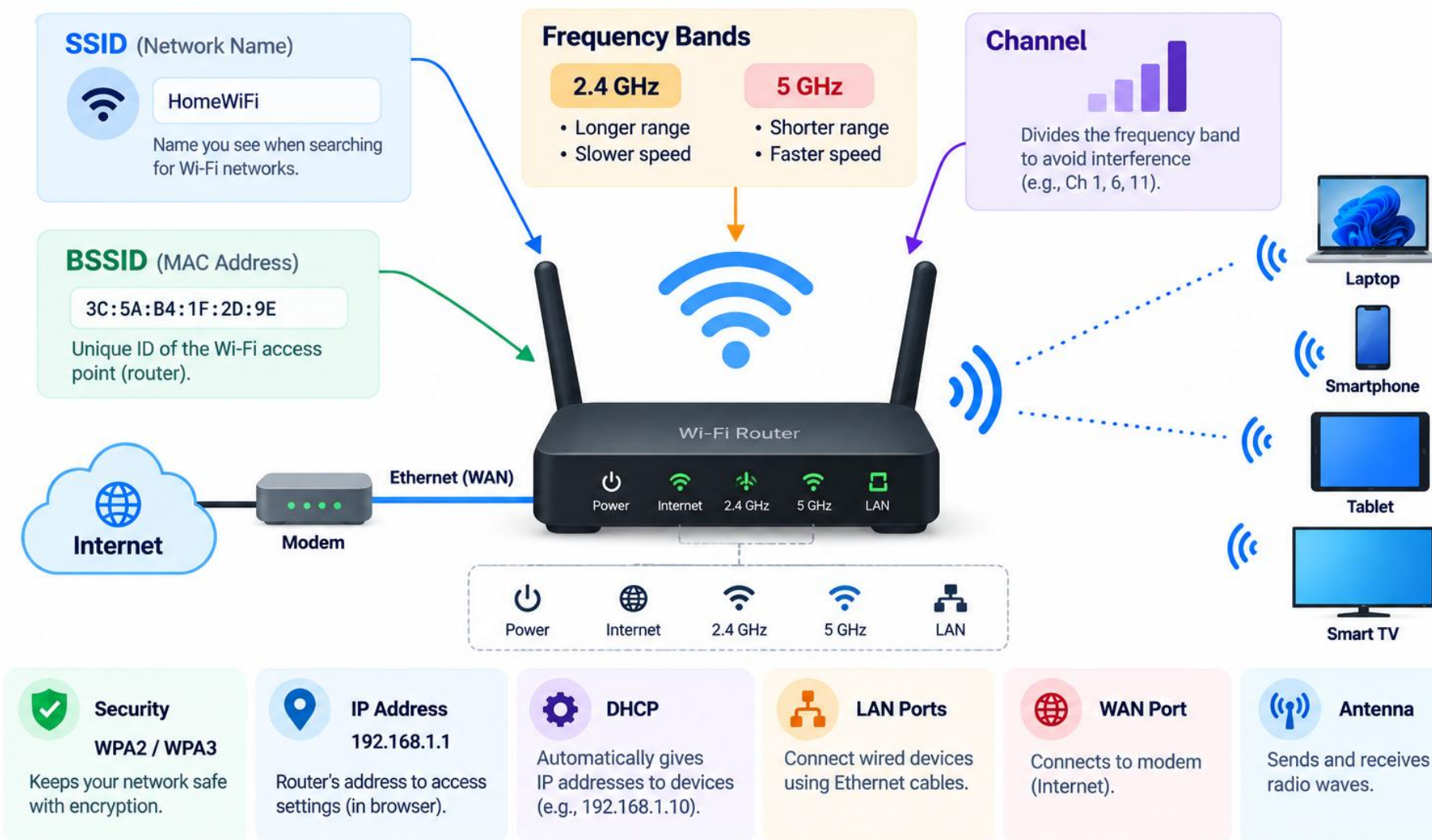
Phishing and Social Engineering Hands-on Lab



<https://samarth-arote.github.io/phishing-simulation-lab/>

Wi-Fi Security

Wi-Fi Security



Public Wi-Fi Risk

Demonstrate

- **wlan show networks mode=ssid**
- **netsh wlan show networks mode=bssid**

Wi-Fi Security



Open Wi-Fi

No password required



Anyone can connect



Less secure



Data can be seen by others



Common in cafés, airports, hotels

Connected



Phone



Laptop



Tablet



Wi-Fi Router



Password Protected Wi-Fi

Requires a password to connect



Only authorised users can connect



Keeps your data safe



Prevents unauthorised access



Used in homes, offices, schools

Connected



Phone



Laptop



Tablet

Wi-Fi Security

WPA2

Wi-Fi Protected Access 2



Uses AES encryption



Commonly used (WPA2-Personal with password)



Widely supported by most devices



Older standard (introduced in 2004)



Still secure if you use a strong password

WPA3

Wi-Fi Protected Access 3



Uses SAE (Simultaneous Authentication of Equals)



Stronger protection against password-guessing attacks



Better resistance to brute force attacks



Newer standard (introduced in 2018)



Requires newer devices for full support



Recommended (when supported)

Wi-Fi Security

1 Use Strong Encryption

Enable WPA2 or WPA3 (in router settings).



2 Use a Strong Password

Set a long, unique Wi-Fi password.



3 Change Default Settings

Change default admin username and password.



4 Keep Firmware Updated

Install the latest router firmware for security fixes.



5 Use a Unique Network Name (SSID)

Avoid default names (like "TP-Link", "D-Link").



Your Home Wi-Fi Safe • Private • Controlled



6 Disable Unused Features

Turn off WPS, remote management, and UPnP if not needed.



7 Create a Guest Network

Use a separate network for guests and IoT devices.



8 Control Connected Devices

Regularly check and remove unknown devices.



9 Use a Secure Location

Keep the router in a central place at home (not outside windows) to avoid signal leakage.



10 Monitor Your Network

Check router logs and device list regularly for suspicious activity.



Public Wi-Fi Risk

Demonstrate

- **Wifi configuration**

Public Wi-Fi Risk

Public Wi-Fi Risk

1 Untrusted Networks
Public Wi-Fi networks (cafés, airports, hotels) can be used by attackers.

You don't know who else is connected!

- Cafe_WiFi
- Airport_FreeWiFi
- Hotel_Guest
- ...

2 Eavesdropping
Attackers can passively listen to the data transmitted over Wi-Fi.

Listening to your data

Website

Emails
Messages
Passwords
...

3 Traffic Interception
Attackers can intercept and even modify the data between you and a website.

Read
Change
Inject
...

Intercepts & modifies data

Website

4 Fake Login Pages
Attackers create fake login pages that look real to steal your credentials.

Looks real but it's fake!

https://example.com

http://example.com

Login

Sign In

5 Malicious Redirects
You may be automatically redirected to harmful websites without your knowledge.

Automatic Redirect

Malicious Website

6 DNS Manipulation
Attackers can tamper with DNS responses and send you to the wrong website.

Request www.bank.com

Correct IP (bank.com)

Fake IP (attacker's site)

DNS Server

Your Bank Website

Fake Website

7 Rogue Access Points
Attackers set up fake Wi-Fi access points to trick users into connecting.

Home_WiFi

Free_WiFi

Legitimate Access Point

Rogue Access Point (set up by attacker)

Fake network looks legitimate!

8 Evil Twin
A fake access point with the same (or similar) name as a legitimate one, tricking you to connect to it.

Office_WiFi

Office_WiFi

Legitimate Access Point

Evil Twin (fake access point)

Looks the same, but it's not!

- Office_WiFi (Real)
- Office_WiFi (Fake)
- ...

Public Wi-Fi Risk

Demonstrate

- **Wi-Fi Evil Twin**
- **Eavesdropping**
- **Traffic Interception**
- **Malicious Redirect**

Thank You



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